

Dane Cross

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in LinkedIn

🌐 Website

Education

- 2022 – present  **Ph.D. Physics** Universitat Autònoma de Barcelona
Thesis title: *Cosmology at High and Low Redshifts*
- 2020 – 2022  **M.Sc. Physics** Indiana University
Thesis Title: *Formation of Star Clusters in Dark Matter Halos*
- 2016 – 2020  **B.Sc. Physics (Astrophysics)** University of California, Santa Cruz
Thesis title: *Examining the Self-Interaction of Dark Matter through Central Cluster Galaxy Offsets.*
-  **B.A. Computer Science** University of California, Santa Cruz

Research Publications (h-index: 5)

- 1 D. Cross and C. Sánchez, “Inverse Galaxy-Galaxy Lensing: Magnification, Intrinsic Alignments and Cosmology,” *arXiv e-prints*, arXiv:2410.00714, arXiv:2410.00714, Oct. 2024.  DOI: 10.48550/arXiv.2410.00714. arXiv: 2410.00714 [astro-ph.CO].
- 2 D. Cross, G. Thoron, T. E. Jeltema, *et al.*, “Examining the self-interaction of dark matter through central cluster galaxy offsets,” vol. 529, no. 1, pp. 52–58, Mar. 2024.  DOI: 10.1093/mnras/stae442. arXiv: 2304.10128 [astro-ph.CO].
- 3 M. Shahbandeh, C. Ashall, P. Hoeflich, *et al.*, “JWST NIRSpec+MIRI Observations of the nearby Type IIP supernova 2022acko,” *arXiv e-prints*, arXiv:2401.14474, arXiv:2401.14474, Jan. 2024.  DOI: 10.48550/arXiv.2401.14474. arXiv: 2401.14474 [astro-ph.HE].
- 4 J. Pearson, D. J. Sand, P. Lundqvist, *et al.*, “Strong Carbon Features and a Red Early Color in the Underluminous Type Ia SN 2022xkq,” vol. 960, no. 1, 29, p. 29, Jan. 2024.  DOI: 10.3847/1538-4357/ad0153. arXiv: 2309.10054 [astro-ph.HE].
- 5 C. Sánchez, A. Alarcon, G. M. Bernstein, *et al.*, “The Dark Energy Survey Year 3 high-redshift sample: selection, characterization, and analysis of galaxy clustering,” vol. 525, no. 3, pp. 3896–3922, Nov. 2023.  DOI: 10.1093/mnras/stad2402. arXiv: 2211.16593 [astro-ph.CO].
- 6 V. Wetzell, T. E. Jeltema, B. Hegland, *et al.*, “Velocity dispersions of clusters in the Dark Energy Survey Y3 redMaPPer catalogue,” vol. 514, no. 4, pp. 4696–4717, Aug. 2022.  DOI: 10.1093/mnras/stac1623. arXiv: 2107.07631 [astro-ph.CO].
- 7 K. Graham, J. O’Donnell, M. M. Silverstein, *et al.*, “Cool Cores in Clusters of Galaxies in the Dark Energy Survey,” *Research Notes of the American Astronomical Society*, vol. 7, no. 5, 86, p. 86, May 2023.  DOI: 10.3847/2515-5172/acd1e9. arXiv: 2305.02365 [astro-ph.CO].

Participation in Conferences and Workshops

- December 2024  **ZTFIa Barcelona**
Collaboration meeting for the type Ia supernova cosmology group within the Zwicky Transient Facility analysis teams.
Gave a talk on the mass vs. velocity approach to identifying systematic effects in $f\sigma_8$ measurements.

Participation in Conferences and Workshops (continued)

- November 2024  *IFAE Pizza Seminar*
A weekly seminar series at the Institut de Física d'Altes Energies where people from the department give interdisciplinary talks on their research.
Gave a talk on Inverse Galaxy-Galaxy Lensing.
- October 2024  *Mock Barcelona*
Invite-only workshop on how to map the galaxy-halo connection.
Gave a talk on Inverse Galaxy-Galaxy Lensing.
-  *Workshop: The Role of SNIa in Current Cosmological Tensions.*
Four-day workshop with the purpose of discussing systematics and unaddressed issues with SNIa in cosmological measurements.
- July 2024  *Reunión Científica de la Sociedad Española de Astronomía.*
Meeting of all Spanish institutions to discuss progress and synergies in the area of Astronomy.
Gave a talk during the Dark Energy Survey Symposium about Inverse Galaxy-Galaxy Lensing.
- June 2024  *Cosmology Summer School at ICTP.* Cosmology summer school on the theoretical foundations of modern cosmology measurements.
Presented a poster on Inverse Galaxy-Galaxy Lensing.
- May 2024  *DES Collaboration Meeting in Barcelona.* Meeting of the DES Collaboration to discuss progress on the science being done with the third and final data release of the program.
Gave an update on the status of the year 6 redshift methodology for the Bayesian Fourier Domain shape and flux catalog.
- November 2023  *DES Key Projects Workshop at Princeton.* Meeting of the DES Collaboration Key Projects group to discuss progress on the 3x2pt cosmology key papers for the final, year 6 data release.
- October 2023  *DES Collaboration Meeting in Urbana-Champaign.* Meeting of the DES Collaboration to discuss progress on the science being done with the third and final data release of the program.
Gave an update on the status of the year 6 redshift methodology for the Bayesian Fourier Domain shape and flux catalog.
- August 2023  *SLAC Summer Institute.* Introduced methods for Artificial Intelligence and Machine Learning and their successful applications across the fundamental physics. This SSI intends to inspire invigorated efforts for new revelations on how the rapidly developing field of Artificial Intelligence can change the ways that data is analyzed in fundamental physics.
Gave poster presentation titled: "Inverse Galaxy-Galaxy Lensing: Magnification, intrinsic alignments and cosmology"

Participation in Conferences and Workshops (continued)

- June 2023  *Michigan Summer School in Cosmology*. Covered topics across the field of cosmology ranging from theory to experiment, but the emphasis will be on computation and the interface between data and theory. An overarching goal of the school will be to provide participants with concrete practical skills for data modeling. The focus will be on the theoretical underpinnings of inflation, dark matter, and dark energy using modern measurements, principally those coming from the LSS and CMB.
- January 2023  *DES Collaboration Meeting in Portsmouth, UK* Meeting of the DES Science Collaboration to discuss developments of analyses for the final weak lensing dataset. Gave two talks: (1) Final Updates for BCG Offsets Paper, (2) Inverse Galaxy-Galaxy Lensing: Magnification, intrinsic alignments and cosmology

Employment History

- May – August 2022  **Software Consultant** Phil Richerme's Quantum Information Lab. Examined and improved the data analysis pipeline for an ion trap experiment.
- 2020 – 2022  **Associate Instructor** Indiana University Physics. Facilitated labs for Introduction to Physics with Algebra, Introduction to Physics with Calculus, and Honors Physics II containing 30 students per section.
- June – August 2022  **Software Consultant** SETI. Overhauled the software analysis code for a laser diffraction (CheMin) instrument which is on the Curiosity Rover. The analysis was the backend code for a user interface which scientists use to analyze the chemical contents of the soil on Mars.
- June – August 2019  **Software Consultant** CERN. Restructured the NA62 analysis code base to be more efficient and user-friendly. Consolidated the interface so the user can run Monte Carlo simulations, access reconstructed data, and create plots in a fraction of the computing time it would have taken previously.
- June – September 2018  **Backend Software Engineering Intern** Automation Anywhere. Implemented a System Diagnostics page for the modules in the software package. Followed through from conception to algorithm design to completion.

Awards

- 2023-2026  **Ajuts Joan Oró per a la contractació de personal investigador predoctoral en formació**, PhD program funding.
- 2020  **Dean's Undergraduate Thesis Award**, University of California, Santa Cruz. "Dean's Awards are granted to fifty of the most excellent undergraduate research theses or projects (ten from each academic division) completed at UCSC during the academic year." Awarded for my undergraduate thesis "Examining the Self-Interaction of Dark Matter through Brightest Cluster Galaxy Offsets."